

When Does a Random-Scale Flight Path Look Lognormal?

A spherical bridge note from OpenSky trajectories

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Abstract

An hour of globally sampled aircraft positions produces a tempting picture: RMS departures from endpoint great-circle arcs are positive, strongly right-skewed, and superficially lognormal. We turn that visual hypothesis into a geometric model. Each deviation is factored as $D = AQ$, where A sets route scale and Q is the RMS shape of an endpoint-pinned bridge. We prove that an independent lognormal scale suppresses the non-Gaussian cumulants of $\log Q$ at explicit rates and give a Wasserstein convergence criterion. For a Brownian bridge, the criterion yields a concrete threshold. Yet the observed log-skewness and kurtosis sharply contradict the resulting single-population prediction. The same theorem that explains a lognormal-looking center therefore diagnoses a distinct near-geodesic boundary population. The result is not a universal lognormal law, but a principled route from visual regularity to a falsifiable mixture model.

A visual regularity with a geometric question

The OpenSky Network makes large-scale ADS-B trajectory research possible [1]. We sampled 60 one-minute global state snapshots, retained aircraft with all 60 positions, required at least 25 km between endpoints, and removed any track with a one-minute jump above 30 km. The final cohort contains 1,780 tracks. These positions are receiver-dependent observations, not a complete census of global aviation.

For aircraft i , write $x_i(t) \in S_R^2$ for its position on the spherical Earth at normalized time $t \in [0, 1]$. The direct reference is the minor endpoint geodesic

$$g_i(s) = \text{Exp}_{x_i(0)}\left(s \text{Log}_{x_i(0)} x_i(1)\right), \quad s \in [0, 1]. \quad (1)$$

Here s is dimensionless progress, and $g_i(s)$ is a position rather than a distance. If $d_{S_R^2}$ is spherical distance in kilometres, define cross-track displacement and its sampled RMS by

$$\eta_i(t) = d_{S_R^2}(x_i(t), g_i([0, 1])), \quad D_i = \left[\frac{1}{60} \sum_{j=1}^{60} \eta_i(t_j)^2 \right]^{1/2}. \quad (2)$$

The median D_i is 16.7 km and the mean is 22.7 km. Figure 1 shows why a lognormal model is initially plausible and why that impression is incomplete.

Classical lognormal modeling begins with $\log(D/d_0) \sim N(\mu, \sigma^2)$ for reference distance $d_0 = 1$ km [2]. That describes a shape but supplies no route geometry. We instead ask: *what mechanism makes a spherical path statistic look lognormal, and what would falsify it?*

Scale, shape, and the bridge factor

Factor the observed departure into a kilometre-valued scale and a dimensionless shape,

$$\eta_i(t) = A_i Z_i(t), \quad D_i = A_i Q_i, \quad Q_i = \left[\int_0^1 Z_i(t)^2 dt \right]^{1/2}, \quad (3)$$

where $Z_i(0) = Z_i(1) = 0$. Intuitively, A_i asks *how large* the route departure is, while Z_i asks *what shape* it takes between fixed endpoints. Across a population, independent densities f_A and f_Q induce the product law

$$f_D(d) = \int_0^\infty f_A(d/q) f_Q(q) \frac{dq}{q}. \quad (4)$$

The Jacobian $1/q$ is the signature of multiplicative rather than additive mixing. A single lognormal is therefore not the starting mechanism; it is a possible approximation to this geometry-driven product.

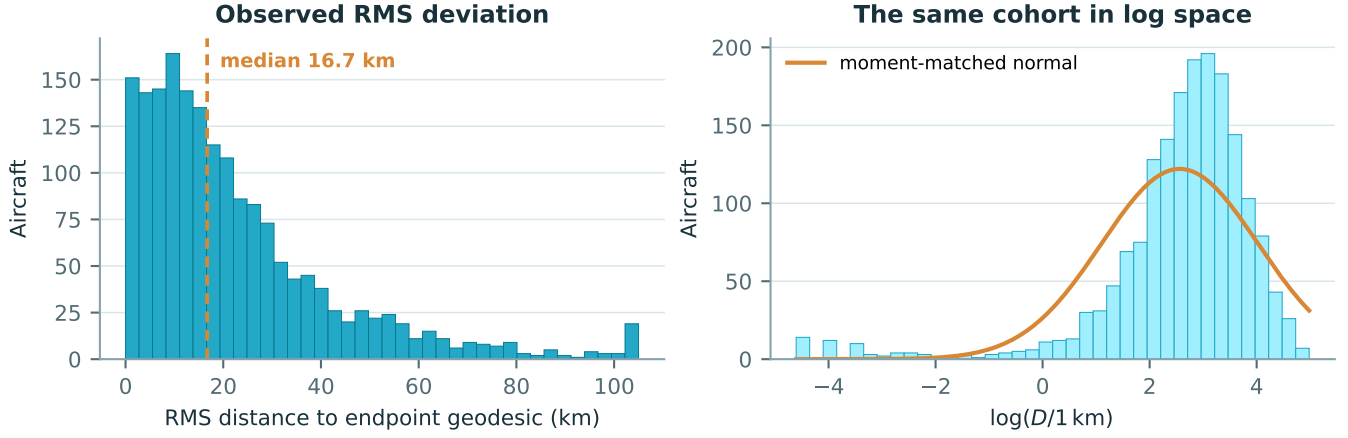


Figure 1: RMS spherical deviation for the quality-filtered cohort. Left: the application histogram, truncated visually at the 99th percentile while retaining tail observations in the end bin. Right: the 1,776 positive values in log space with a moment-matched Gaussian. The broad center looks compatible with lognormality, but the long lower tail does not.

When the product becomes lognormal

Set $U_\ell = \log(A_\ell/d_0)$ and $V_\ell = \log Q_\ell$ for a sequence of scale-shape populations. The index ℓ labels increasing scale dominance, not time or aircraft.

Theorem 1 (Lognormal smoothing). *Suppose $U_\ell \sim N(\mu_\ell, \sigma_\ell^2)$ is independent of V_ℓ , with $m_\ell = \mathbb{E}V_\ell$ and $\tau_\ell^2 = \text{Var}(V_\ell) < \infty$. Define*

$$W_\ell = \frac{\log(D_\ell/d_0) - (\mu_\ell + m_\ell)}{\sqrt{\sigma_\ell^2 + \tau_\ell^2}}.$$

If $\tau_\ell^2/(\sigma_\ell^2 + \tau_\ell^2) \rightarrow 0$, then $W_\ell \Rightarrow N(0, 1)$. More precisely, for $G_\ell \sim N(m_\ell, \tau_\ell^2)$,

$$d_{W_2}(W_\ell, N(0, 1)) \leq \frac{d_{W_2}(V_\ell, G_\ell)}{\sqrt{\sigma_\ell^2 + \tau_\ell^2}},$$

where d_{W_2} is quadratic Wasserstein distance.

Proof. Replace V_ℓ by its Gaussian moment match G_ℓ . Then $U_\ell + G_\ell$ is exactly Gaussian with the displayed centering and variance. Convolution with the same independent U_ℓ contracts W_2 . Finally, independent coupling gives $d_{W_2}(V_\ell, G_\ell) \leq \sqrt{2}\tau_\ell$, so the stated variance ratio forces convergence. $\square$

The theorem formalizes a simple picture: if fleet-to-fleet variation in route scale is broad enough, it blurs the finer distributional signature of route shape.

Proposition 1 (Exact cumulant attenuation). *For one fixed population, let $\rho = \tau_Q^2/(\sigma_A^2 + \tau_Q^2)$, where $\sigma_A^2 = \text{Var}(\log(A/d_0))$ and $\tau_Q^2 = \text{Var}(\log Q)$. For every $r \geq 3$,*

$$\kappa_r(W) = \frac{\kappa_r(\log Q)}{(\sigma_A^2 + \tau_Q^2)^{r/2}}.$$

Consequently,

$$\text{Skew}(W) = \text{Skew}(\log Q)\rho^{3/2}, \quad \text{ExKurt}(W) = \text{ExKurt}(\log Q)\rho^2.$$

Proof. Independent cumulants add, and a Gaussian has no cumulants above order two. Standardizing by the total log-variance gives the first identity; writing the third and fourth cumulants in standardized form gives the second line. $\square$

These identities turn “approximately lognormal” into a checkable statement: shape-induced skewness decays as $\rho^{3/2}$ and excess kurtosis as ρ^2 .

The Brownian benchmark and its failure

A standard Brownian bridge is the simplest endpoint-pinned random shape. Its Karhunen–Loève expansion and RMS functional are

$$Z(t) = \sum_{k=1}^{\infty} \frac{\sqrt{2}}{\pi k} \xi_k \sin(\pi k t), \quad Q^2 = \int_0^1 Z(t)^2 dt = \sum_{k=1}^{\infty} \frac{\xi_k^2}{\pi^2 k^2}, \quad (5)$$

with independent $\xi_k \sim N(0, 1)$. This is the classical quadratic Brownian-bridge functional underlying Cramér–von Mises theory [3]; its Laplace transform is

$$\mathbb{E}e^{-\lambda Q^2} = \left(\frac{\sqrt{2\lambda}}{\sinh \sqrt{2\lambda}} \right)^{1/2}, \quad \lambda > 0. \quad (6)$$

A two-million-draw, 500-mode numerical evaluation gives

$$\text{SD}(\log Q) \approx 0.385, \quad \text{Skew}(\log Q) \approx 0.178, \quad \text{ExKurt}(\log Q) \approx -0.348.$$

Substitution in the proposition yields a concrete conclusion: $\sigma_A \gtrsim 0.50$ pushes both surviving standardized cumulants below 0.05 in magnitude. A moderately broad lognormal scale should therefore make this Brownian product look very nearly lognormal.

The observed cohort supplies a sharp diagnostic. Among the 1,776 positive D_i values,

$$\text{SD}(\log(D/d_0)) = 1.465, \quad \text{Skew} = -2.40, \quad \text{ExKurt} = 8.11.$$

Variance matching under the independent Brownian model implies $\sigma_A \approx 1.413$. At that scale spread, the proposition predicts skewness about 0.003 and excess kurtosis about -0.002 —essentially Gaussian in log space. The data are nowhere close. The mismatch is driven by a near-geodesic boundary: 53 tracks have recorded RMS deviation at or below 0.1 km.

From an attractive fit to a useful rejection

The mathematical arc reverses the usual modeling order. We did not choose a flexible mixture merely because a histogram fit was imperfect. Geometry first proposed $D = AQ$; the smoothing theorem then showed that the Brownian single-regime model should be *more* Gaussian in log space than the data; only then did a mixture become necessary.

A natural next model is

$$f_D(d) = \pi_0 f_0(d) + \sum_{c=1}^C \pi_c f_D(d | c), \quad \pi_c \geq 0, \quad \sum_{c=0}^C \pi_c = 1, \quad (7)$$

where f_0 represents the near-geodesic boundary and the continuous components represent distinct route-shape or flight-phase regimes. The current hour cannot identify those classes, and endpoint geodesics are geometric counterfactuals rather than operationally feasible routes. Those limitations are substantive. Still, the note establishes a clean diagnostic principle: broad lognormal scaling can hide ordinary bridge-shape nonnormality, so a large residual departure is evidence for heterogeneity, dependence, or a boundary mechanism—not merely a slightly misspecified bridge.

References

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